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A structural reformulation of the local entropy inequality based on the thermodynamic admissibility of exchanged energy

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ABSTRACT

The local entropy inequality in continuum thermodynamics is conventionally formulated by associating heat exchange with entropy transfer through the Clausius relation. This formulation, while appropriate for conventional thermal interactions, does not by itself establish that every energy-transfer contribution appearing in an internal-energy balance constitutes Clausius-effective heat. The present work addresses this unresolved thermodynamic attribution by introducing a structural decomposition of exchanged energy into Clausius-effective and non-Clausius contributions. The proposed framework establishes a necessary and sufficient criterion: an exchanged-energy contribution enters the local entropy-transfer term if and only if it admits a reversible thermodynamic representation consistent with the Clausius relation. Accordingly, the total energy balance remains unchanged and retains all energetic mechanisms, whereas the entropy-transfer term is restricted to thermodynamically admissible Clausius-effective exchanges. Non-Clausius contributions—including mechanically mediated dissipation, irreversible deformation, and internal structural energy conversion—remain fully represented in the energy balance and may contribute to internal entropy production through the irreversible evolution of the system, without being identified as Clausius heat. A general identification procedure is formulated based on the separation of thermal and non-thermal energy pathways, reversible and irreversible mechanical contributions, and independent and constrained energetic components. The framework is further connected to the structural viewpoint of Energy Structure Theory and General Energy Behavior (EST-GEB), providing a broader basis for classifying energy-transfer pathways across complex continua. For viscous heat-conducting fluids, the formulation distinguishes Fourier heat transfer from viscous dissipation without conflating a volumetric dissipative power with a heat-flux contribution. The resulting framework preserves the first law and the second law while sharpening the thermodynamic attribution of exchanged energy and providing a systematic basis for extending local entropy analysis to strongly coupled, multiphysics, and nonequilibrium systems.

1. Introduction

Entropy occupies a central role in thermodynamics, serving as the cornerstone of the second law and as a measure of irreversibility in physical processes [1,2]. In classical thermodynamics, entropy is introduced through the Clausius relation, in which entropy change is defined via reversible heat exchange divided by absolute temperature [3–6]. This construction emphasizes a crucial conceptual point: Entropy is not associated with arbitrary energy transfer, but only with that portion of energy exchange that admits a Clausius interpretation. Notably, from the perspective of energy structure evolution laws [7–11], only independent energy components with dynamically relevant time derivatives contribute to entropy production, highlighting the selective nature of Clausius-effective heat.

Over the past several decades, considerable effort has been devoted to extending the classical formulation of the second law to increasingly complex materials and irreversible processes [12–14]. Modern developments in continuum thermodynamics, extended irreversible thermodynamics, and generalized continuum theories have emphasized that entropy production often involves multiple interacting mechanisms rather than a single thermal pathway [15–18]. These developments have substantially broadened the applicability of thermodynamic principles while preserving the local balance structure inherited from the classical Clausius-Duhem framework [19,20].

Representative examples include the framework of Extended Irreversible Thermodynamics, GENERIC formulations, internal-variable theories, and generalized continuum models [15–17], all of which distinguish different sources of dissipation and irreversible evolution.

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Nevertheless, these approaches generally retain the conventional assumption that the heat quantities appearing in the local energy balance also constitute the thermal contributions entering the entropy inequality [19]. Consequently, although the origin of irreversibility is treated in increasingly sophisticated ways, the thermodynamic interpretation of the heat terms themselves remains largely unchanged.

Recent advances in non-equilibrium thermodynamics emphasize the necessity of revisiting foundational inequalities to accommodate complex dissipative behaviors. This quest for a generalized and unified framework for entropy production, as highlighted in recent works [9], further reinforces the need for a precise structural decomposition of energy transfer, which constitutes the primary focus of the present study.

From this perspective, the unresolved issue is not the existence of irreversible energy transfer itself, but rather whether every energy contribution formally appearing as heat in the internal-energy balance necessarily satisfies the thermodynamic requirements associated with Clausius heat. Despite extensive developments in non-equilibrium thermodynamics, relatively little attention has been devoted to explicitly separating the thermodynamic admissibility of heat exchange from its energetic representation in the balance equations. This conceptual gap provides the principal motivation for the structural reformulation presented in the following sections.

Recent studies on entropy generation in complex and rheological continua have further emphasized the importance of carefully characterizing thermal and solutal energy transport in non-equilibrium systems. In particular, entropy-optimized analyses of Jeffrey nanofluids incorporating Cattaneo–Christov heat-flux theory have examined the consequences of finite-speed thermal transport and rheological effects on entropy generation. Related investigations of Reiner-Rivlin materials have considered modified thermal and solutal fluxes in convective transport, while studies of double-diffusive rheological nanomaterials have addressed entropy generation under non-Fourier Cattaneo-Christov transport conditions relevant to thermal energy-storage systems. Neural-network-based approaches have also been employed to analyze entropy generation in radiative rheological flows. These studies demonstrate the increasing use of generalized heat-flux models, rheological descriptions, and computational tools to characterize entropy generation in complex continua [21–24]. However, such formulations generally focus on quantifying entropy generation once the relevant energy and heat-transfer mechanisms have been specified. The present work addresses a more fundamental structural question: whether every energy-transfer contribution appearing in an internal-energy balance should automatically be regarded as a Clausius-effective heat contribution to the local entropy inequality.

1.1. Novelty and distinction from previous work

The novelty of the present study lies in addressing a question that is logically prior to the conventional calculation of entropy generation: whether every energy-transfer contribution appearing in a continuum energy balance is thermodynamically admissible as Clausius heat in the local entropy inequality. Previous studies have extensively developed constitutive models, generalized heat-flux relations, rheological descriptions, and computational approaches for evaluating entropy generation in complex continua [21–24]. These contributions are essential for quantifying irreversible behavior once the relevant energy-transfer mechanisms have been specified. The present work addresses a different level of the problem by introducing a structural criterion for classifying exchanged energy according to its admissibility for Clausius entropy transfer.

Specifically, the proposed framework (i) separates Clausius-effective and non-Clausius energy contributions without modifying the first-law energy balance, (ii) identifies the existence of a reversible thermodynamic representation as the criterion for explicit participation in the Clausius entropy-transfer term, (iii) distinguishes boundary entropy transfer from internal entropy production, and (iv) provides a general identification procedure applicable, in principle, to mechanically, structurally, chemically, and electromagnetically coupled continua. The viscous-fluid example is then used to demonstrate the distinction between thermal boundary heat transfer and mechanically generated viscous dissipation. Thus, the contribution of the present study is not another constitutive entropy-generation model, but a thermodynamic classification framework that determines which energy-transfer pathways may consistently be represented as Clausius-effective heat before entropy-transfer calculations are performed.

In continuum thermodynamics, however, the entropy balance is typically obtained by combining the local form of the energy balance with the Clausius-Duhem inequality [25,26]. In this procedure, the heat flux and volumetric heat supply appearing in the energy balance are directly inserted into the entropy inequality, implicitly identifying them with Clausius heat [27,28]. This identification is rarely stated explicitly, yet it is essential for the validity of the derivation.

Classical continuum thermodynamics established the Clausius–Duhem inequality as the principal local statement of the second law, providing the foundation for constitutive modeling in thermoelastic and dissipative materials. Subsequent developments introduced generalized continuum descriptions, internal-variable theories, and non-equilibrium thermodynamic frameworks capable of representing increasingly complex irreversible phenomena. More recent studies have also examined entropy production from statistical and information-theoretic viewpoints, emphasizing the microscopic origin of irreversibility and the multiplicity of dissipation mechanisms. Despite these advances, the local entropy inequality continues to employ essentially the same interpretation of heat exchange inherited from the classical formulation.

The purpose of this paper is to show that this identification is not generally justified. Modern perspectives on entropy, irreversibility, and dissipation indicate that a continuum may exchange energy in ways that formally appear as heat in the energy balance, but do not correspond to Clausius heat [7,9,29–34]. When such energy transfers are indiscriminately included in the entropy inequality, a conceptual inconsistency arises between the foundations of thermodynamics and their application in continuum mechanics. In fact, the classical formulation remains valid only for a narrow class of materials - primarily simple thermoelastic continua with Fourier heat conduction - where all heat-like contributions happen to coincide with thermodynamically admissible heat. The present study explicitly revises the local entropy inequality of continuum mechanics by structurally separating Clausius-effective heat from non-Clausius energy transfer.

This paper argues that this inconsistency has remained largely unnoticed because the classical formulation works correctly for a restricted class of materials and processes. However, as continuum theories are extended to encompass complex materials, strong couplings, internal dissipation mechanisms, and non-equilibrium effects, the need for a more precise distinction becomes unavoidable.

The central contribution of this work is to show that only a restricted portion of the energy exchanged with a continuum - namely, the part that admits a reversible thermodynamic interpretation - contributes to entropy in the Clausius sense. By examining the classical derivation of the local entropy inequality, we demonstrate that the standard formulation implicitly treats all heat-like terms appearing in the energy balance as Clausius heat. This implicit step, although rarely acknowledged, is the root of a structural inconsistency that arises when mechanically mediated or non-thermal energy transfers are present. Building on this observation, the present study introduces a revised entropy inequality in which only the Clausius-effective part of heat exchange enters the second law. This selective formulation restores consistency between the classical definition of entropy and its continuum-mechanical application.

To visually summarize the conceptual distinction between Clausius-effective and non-Clausius heat introduced in this work, Fig. 1 provides a

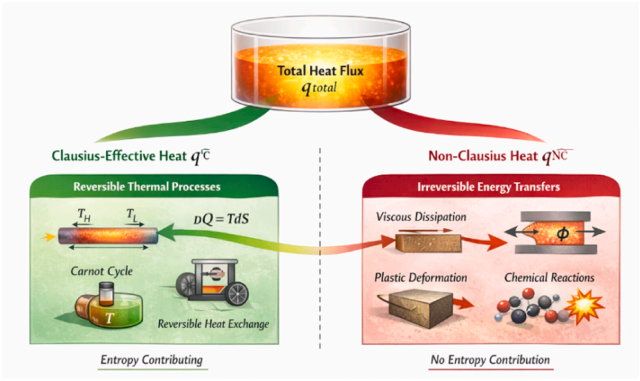


Fig. 1. Schematic illustration of the decomposition of total heat into Clausius-effective and non-Clausius components. Clausius-effective thermal exchanges (green) contribute directly to the Clausius entropy-transfer term, whereas non-Clausius energy transfers (red) are not represented as Clausius entropy-transfer contributions and may nevertheless participate in irreversible entropy production through the complete thermodynamic evolution. This figure conceptually summarizes the distinction that underlies the revised entropy inequality.

schematic representation of the total heat flux in a continuum system. The central container represents the total heat $\boldsymbol{q}_{total}$ entering the system, which is then divided into two pathways. On the left, Clausius-effective heat $\boldsymbol{q}^{C}$ is illustrated in green, highlighting reversible thermal processes such as idealized heat conduction, Carnot-like cycles, or other interactions that can, in principle, be reversed and contribute directly to entropy production. On the right, non-Clausius heat $\boldsymbol{q}^{NC}$ is depicted in red, representing energy transfers that do not admit a reversible thermodynamic interpretation. Examples include viscous dissipation, plastic deformation, and chemical reactions whose energy changes cannot be associated with a reversible thermal reservoir.

The present work does not claim that the classical entropy inequality is universally incorrect in all continuum systems. Rather, we show that its standard form implicitly assumes that every contribution appearing as heat in the energy balance is thermodynamically Clausius-effective. This assumption is valid only when the exchanged energy admits a reversible thermal interpretation. In more general coupled systems, however, some energy transfers enter the internal-energy balance without corresponding to a Clausius-type heat interaction. The purpose of the proposed formulation is therefore not to reject the second law, but to identify the precise structural conditions under which the classical entropy inequality remains valid and to separate those contributions that genuinely belong to entropy production from those that do not.

The Fig. 1 conveys not only the conceptual decomposition of total heat but also emphasizes that only the Clausius-effective portion is relevant to the entropy balance, while the non-Clausius fraction is not represented as a Clausius entropy-transfer contribution. By placing this figure here, at the conclusion of the introduction, the reader gains an immediate, intuitive grasp of the distinction that underpins the subsequent formulation of the modified entropy inequality, making the abstract conceptual framework more tangible and preparing the ground for the technical derivations in Section 4.

1.2. *Physical interpretation of Fig. 1*

Fig. 1 should be interpreted as a structural thermodynamic map rather than as a quantitative partition of energy. The two pathways shown in the schematic represent different physical modes by which energy can enter, leave, or be redistributed within a continuum. The Clausius-effective pathway corresponds to an energy exchange that can be associated with a reversible thermal reference process and can therefore be represented in the entropy-transfer term. The non-Clausius pathway represents energy transfer or conversion for which such a reversible thermal representation is not established. Importantly, the latter designation does not imply zero entropy production. An internally dissipative mechanism may generate entropy while remaining absent from the Clausius entropy-transfer term. Thus, the physical distinction illustrated in Fig. 1 is between the mode of energy transfer and its thermodynamic representation in the entropy balance, rather than between "entropy-producing" and "entropy-free" energy. This distinction is essential for interpreting the subsequent local formulation, particularly in systems where mechanical work, viscous dissipation, structural evolution, and thermal exchange occur simultaneously.

Taken together, these considerations indicate that the conventional identification of all heat-like energy transfers with Clausius heat obscures the true thermodynamic structure of irreversible processes. A more selective interpretation of heat, one that respects the distinction between reversible thermal exchange and energetically significant but non-Clausius mechanisms, reveals a deeper consistency between continuum mechanics and the foundational principles of thermodynamics. By revisiting the role of heat within the entropy balance, we establish a framework in which entropy production is attributed only to thermodynamically meaningful pathways, while allowing mechanically mediated or structurally constrained energy flows to influence the evolution of the system without being misclassified as contributors to entropy. This refined viewpoint opens the way to a more transparent and structurally coherent formulation of the local entropy inequality, especially in complex or strongly coupled continua where conventional assumptions no longer suffice.

The central question examined in this paper is not introduced merely as a consequence of energy structure evolution laws. Rather, it emerges from a longstanding conceptual difficulty in non-equilibrium thermodynamics of continua: the common identification of all heat terms in the energy balance with entropy-producing Clausius heat. This identification is often adopted implicitly in standard local formulations, yet its validity becomes less evident in the presence of strong internal coupling, viscous dissipation, structural evolution, or other non-classical energy-transfer mechanisms. In this sense, the present work is rooted primarily in the tradition of continuum non-equilibrium thermodynamics and in the thermomechanical analysis of irreversible processes, while EST and GEB are invoked only as supporting structural viewpoints that help interpret the proposed distinction.

The motivation of the present work should therefore be understood within the broader tradition of non-equilibrium thermodynamics and statistical physics, where the relation between microscopic irreversibility and macroscopic entropy production has been studied for decades. In that setting, heat, dissipation, and internal structural evolution are not treated as purely formal terms, but as manifestations of underlying energy-transfer mechanisms whose thermodynamic significance depends on their dynamical role and admissibility within the entropy principle. The present distinction between Clausius-effective and non-Clausius contributions is thus intended as a continuum-level structural interpretation of a familiar problem in irreversible thermodynamics: how to separate those energy exchanges that genuinely contribute to entropy production from those that merely appear in the energy balance without admitting a reversible Clausius interpretation.

The present study develops a structural reformulation of the local entropy inequality based on the distinction between Clausius-effective and non-Clausius energy transfer. A rigorous thermodynamic admissibility criterion is established, demonstrating that exchanged energy contributes explicitly to the entropy balance if and only if it admits a reversible Clausius representation. This criterion provides the theoretical basis for the proposed decomposition and leads naturally to a revised local entropy inequality in which only Clausius-effective energy contributes directly to entropy transport. The resulting formulation preserves the classical Clausius-Duhem inequality as a limiting case while extending its applicability to systems involving energy-transfer mechanisms that cannot be consistently interpreted as reversible heat exchange.

Accordingly, the discussion that follows is framed not only as a reformulation of the entropy inequality, but also as a continuum-level interpretation of irreversibility in the sense developed by non-equilibrium thermodynamics and statistical mechanics.

The present paper is organized as follows. First, we identify the structural basis of Clausius-effective heat and its relation to reversible thermodynamic interpretation. Next, we examine the conventional local entropy inequality and isolate the assumption that all heat-like energy transfers must contribute to entropy production. We then formulate a revised entropy inequality by decomposing total exchanged energy into Clausius-effective and non-Clausius parts. Finally, we illustrate the framework through a viscous-fluid example and discuss its compatibility with broader thermodynamic formulations.

2. Clausius-effective heat: structural basis and thermodynamic meaning

2.1. Clausius-effective heat

The classical definition of entropy originates from the work of Carnot and Clausius, who established that for a reversible process the differential of entropy is given by [1]:

$$dS = \frac{\delta Q_{rev}}{T} \tag{1}$$

This expression already contains a strong restriction. In fact, only reversible heat exchange contributes to entropy change. For irreversible processes, Clausius introduced the inequality as follows [1]:

$$dS \geq \frac{\delta Q}{T} \tag{2}$$

Which again presupposes that the heat δQ admits a thermodynamic interpretation compatible with reversible reference processes.

Modern treatments of thermodynamics emphasize that entropy is a state function reflecting the irreversible redistribution of energy among microscopic degrees of freedom [35–39]. From this viewpoint, not every energy transfer qualifies as heat in the Clausius sense. Energy may be transferred in ways that are mechanically mediated, structurally constrained, or internally dissipated without a corresponding reversible heat counterpart.

Several authoritative sources have highlighted this limitation, noting that the concept of heat is not primitive but context-dependent [39–43]. In particular, energy transfers associated with internal friction, inelastic deformation, chemical reactions, or active processes may appear as thermal energy in a balance equation, yet lack a direct Clausius interpretation. Only the portion of energy exchange that can be associated with a hypothetical reversible thermal interaction at a well-defined temperature contributes to entropy change.

This distinction implies that, in general, the total heat appearing in an energy balance may be decomposed into two parts: a Clausius-effective heat that contributes to entropy, and a remainder that does not. Ignoring this distinction amounts to extending the scope of the Clausius inequality beyond its domain of validity.

2.2. Structural domain of entropy and its clausius basis

Within the energy-structure representation, the total internal energy may be expressed in terms of dynamically independent (e_i^{ind}) and dependent (e_j^{dep}) components. Independent components provide natural candidates for identifying distinct reversible thermodynamic exchange pathways because their evolution can be associated with independently evolving energetic degrees of freedom. However, thermodynamic independence alone does not imply that a component is necessarily entropy-producing or entropy-neutral. Likewise, dependent or constrained components cannot be universally excluded from entropy production. Their thermodynamic role must instead be determined from the constitutive relations, coupling mechanisms, and constraints governing the particular system. Accordingly, the independent/dependent decomposition is used here as a structural aid for identifying candidate Clausius-effective pathways rather than as an independent criterion that, by itself, determines entropy production.

The existence of such implicit assumptions regarding the thermodynamic nature of heat-like terms has also been noted in classical and non-classical continuum formulations [37], who emphasize the need for thermodynamically consistent treatment of energy exchanges in generalized continuum theories.

2.3. The entropy inequality in continuum thermodynamics: a conceptual inconsistency

In the following analysis, all balance laws are stated explicitly before the entropy inequality is revised, so that the proposed modification is derived from a transparent thermodynamic structure rather than introduced by definition.

Before introducing the proposed reformulation, we first state the standard local balance relations used as the governing framework for the present analysis. The local balance of internal energy and the local entropy inequality adopted here are those of classical continuum thermodynamics and rational thermodynamics, in which the first law is expressed through the internal-energy balance and the second law is expressed through a local entropy inequality [12,19,25–27]. These relations are used here as the established thermodynamic baseline rather than as newly proposed governing equations. The purpose of the present work is to reconsider the thermodynamic admissibility of the energy-transfer terms appearing in these relations and to distinguish their entropy-carrying and non-Clausius contributions.

The governing equations of the continuum flow are stated in vector form to establish the mechanical and energetic baseline of the analysis. For a continuum with mass density ρ, velocity field $\boldsymbol{V}$, Cauchy stress tensor $\boldsymbol{\sigma}$, body-force density $\boldsymbol{b}$, specific internal energy e, heat flux $\boldsymbol{q}$, and volumetric heat supply r, the local governing relations may be written as [25]:

$$\frac{\partial \rho}{\partial t} + \nabla.(\rho \boldsymbol{V}) = 0 \tag{3-a}$$

$$\rho \frac{D\boldsymbol{V}}{Dt} = \nabla.\boldsymbol{\sigma} + \rho \boldsymbol{b} \tag{3-b}$$

And finally:

$$\rho \dot{e} = \boldsymbol{\sigma} : \nabla \boldsymbol{V} - \nabla.\boldsymbol{q} + r \tag{3-c}$$

The first relation represents conservation of mass, the second represents linear momentum balance, and the third represents the local internal-energy balance. These equations are stated here in vector and tensor form to make explicit the complete continuum-mechanical framework within which the subsequent entropy analysis is performed. The present work does not modify these governing equations. Instead, it examines the thermodynamic admissibility of the energy-transfer terms appearing in the internal-energy balance and their representation in the local entropy inequality.

For the viscous-fluid example considered later, the constitutive specialization is introduced only after the general thermodynamic formulation has been established. In particular, the Newtonian stress relation and Fourier heat-flux relation are treated as constitutive specifications rather than as modifications of the governing balance laws.

Also, the local entropy inequality is commonly expressed as [25]:

$$\rho \dot{S} \geq -\nabla.\left(\frac{\boldsymbol{q}}{T}\right) + \frac{r}{T} \tag{4}$$

This inequality is usually justified by appealing to the Clausius-Duhem argument in classical continuum mechanics [12–15]. A critical

observation is that the quantities $\boldsymbol{q}$ and r entering the entropy inequality are taken directly from the energy balance, without any additional qualification. This procedure implicitly assumes that the energy transfer represented by $\boldsymbol{q}$ and r is entirely Clausius heat. However, as discussed in the previous section, this assumption is not generally valid.

The inconsistency can now be stated precisely. Based on that, the entropy inequality of continuum mechanics applies the Clausius concept of heat to energy transfers whose thermodynamic nature has not been established. As a result, the inequality may attribute entropy production to energy exchanges that do not possess a Clausius interpretation. This issue does not typically manifest in simple thermoelastic materials with Fourier heat conduction, which explains why the classical framework has been remarkably successful. Nevertheless, once more general materials and processes are considered, the conceptual gap becomes evident and calls for a revision of the formulation.

3. Selective clausius heat and a revised entropy inequality

The analysis in the previous sections shows that the standard entropy inequality of continuum mechanics implicitly treats all heat appearing in the energy balance as Clausius heat. However, classical and modern thermodynamics indicate that this identification is not generally justified. In this section, we propose a structural revision of the entropy inequality that restores consistency with the Clausius concept of entropy.

In the present framework, the Clausius-effective/non-Clausius distinction is not introduced as an arbitrary splitting rule, but as a structural classification based on whether a given energy transfer can be associated with a reversible thermal pathway and with dynamically independent energy content. A contribution is Clausius-effective only if it can be consistently represented in the entropy balance through a reversible thermodynamic interpretation. By contrast, energy exchanges that are mediated through internal mechanical, structural, or other dependent processes remain admissible in the energy balance but do not enter the entropy inequality as heat.

Axiom 1. (Entropy-transfer axiom)

The transfer of entropy across the boundary of a thermodynamic system is associated exclusively with the portion of energy exchange that carries entropy through the system boundary.

This entropy-carrying contribution is termed Clausius-effective heat, denoted by

$$\delta Q^{C}$$

All remaining energetic contributions that do not directly transport entropy across the system boundary are classified as non-Clausius heat, denoted by

$$\delta Q^{NC}$$

Accordingly, the total heat interaction admits the decomposition

$$\delta \mathrm{Q} = \delta Q^{C} + \delta Q^{NC}$$

Within the present formulation, only the Clausius-effective component contributes to entropy transfer, whereas the non-Clausius component may influence entropy solely through irreversible internal entropy production.

Remark

The above axiom does not introduce a new physical law.

Rather, it establishes the structural principle upon which the subsequent reformulation of the entropy inequality is developed.

The second law itself remains unchanged; only the thermodynamic attribution of entropy transfer is reformulated.

Corollary 1

As an immediate consequence of Axiom 1, the entropy transfer across the system boundary is expressed exclusively by:

$$\delta S_e = \delta Q^{c}/T$$

Whereas entropy associated with irreversible non-Clausius processes is not represented as a Clausius entropy-transfer term across the boundary. Such processes may, however, contribute to internal entropy production through the complete thermodynamic evolution of the system.

3.1. Decomposition of exchanged heat

The analysis is began by postulating that the total heat exchange appearing in the energy balance can, in general, be decomposed into two distinct parts:

$$\boldsymbol{q} = \boldsymbol{q}^{C} + \boldsymbol{q}^{NC} \tag{5-a}$$

$$r = r^{C} + r^{NC} \tag{5-b}$$

Where $\boldsymbol{q}^{C}$ and r^{C} denote the Clausius-effective heat flux and heat supply, i.e., the part of heat exchange that admits a reversible thermodynamic interpretation. Also, $\boldsymbol{q}^{NC}$ and r^{NC} denote the non-Clausius heat components, which represent energy transfer that does not possess a Clausius interpretation.

This decomposition is not introduced as a constitutive assumption but as a structural distinction motivated by the definition of entropy itself. Entropy is defined through reversible heat exchange. Therefore, only the Clausius-effective part of heat can legitimately enter the entropy balance.

Therefore, the criterion is not experimental labeling in isolation, but consistency with the reversible thermal structure underlying entropy definition and with the decomposition of internal energy into independent and dependent components.

Importantly, the total energy balance remains unchanged. The decomposition does not modify the first law; it only clarifies how the second law should be applied.

Accordingly, the point of departure from the classical formulation is not the second law itself, but the unqualified identification of all heat-like energy transfers with entropy-producing heat exchange. This distinction becomes essential in strongly coupled continua where mechanical, structural, or internal energy exchanges do not admit a reversible Clausius pathway.

3.2. Revised local entropy inequality

With the above distinction, we propose that the local form of the entropy inequality in continuum mechanics should be written as follows:

$$\rho \dot{S} \geq -\nabla.\left(\frac{\boldsymbol{q}^{C}}{T}\right) + \frac{r^{C}}{T} \tag{6}$$

It is important to emphasize that the present formulation does not deny the existence of irreversibility or internal dissipation in continuum systems. Rather, it clarifies that entropy is not a bookkeeping device for all forms of dissipated energy. Dissipative mechanisms may contribute to the total energy balance and affect the system's evolution, but only energy transfers that admit a reversible thermodynamic reference process can be meaningfully associated with entropy in the Clausius sense. The revised inequality therefore refines, rather than weakens, the second law by restricting its application to thermodynamically well-defined heat exchange.

In contrast to the classical formulation, the non-Clausius heat components $\boldsymbol{q}^{NC}$ and r^{NC} do not appear in the entropy inequality. This exclusion is not arbitrary. It follows directly from the requirement that entropy production be associated only with energy transfers that admit a Clausius interpretation.

The revised inequality preserves the mathematical structure of the classical Clausius–Duhem form, while correcting its physical content. It

ensures that entropy production is not artificially attributed to energy exchanges that are thermodynamically ill-defined.

An immediate consequence of the revised entropy inequality is its consistency with the classical Clausius–Duhem formulation. When all exchanged energy satisfies the thermodynamic requirements of Clausius heat, the non-Clausius contributions vanish identically, and the proposed entropy inequality reduces directly to its classical counterpart. The revised formulation therefore constitutes a strict extension of the conventional theory rather than an alternative thermodynamic principle.

Consequently, the proposed framework preserves all predictions of the classical entropy inequality for conventional thermoelastic continua while extending its applicability to systems in which only a subset of the exchanged energy is thermodynamically admissible as Clausius heat. The classical formulation is therefore recovered as a limiting case within the broader theoretical structure introduced here.

Theorem 1. (Necessity of Clausius-Effective Heat in the Local Entropy Inequality).

Let the total heat flux $\boldsymbol{q}$ and volumetric energy supply r be decomposed into Clausius-effective and non-Clausius contributions according to Eq. (5). Assume that the local entropy inequality is required to preserve the classical Clausius interpretation of entropy transfer, namely that every entropy flux contribution must be representable by reversible heat exchange divided by the absolute temperature. Then only the Clausius-effective components may appear explicitly in the entropy flux and entropy supply terms. Any non-Clausius energy contribution cannot consistently enter the local entropy inequality through the classical Clausius relation.

Proof.

By definition, the Clausius entropy transfer is associated exclusively with reversible heat exchange and is locally represented by the quantity

$$dS = \frac{\delta Q_{rev}}{T}$$

Consequently, every term contributing to the entropy flux or entropy supply must admit a reversible thermodynamic interpretation.

Consider an arbitrary energy contribution classified as non-Clausius. By construction, such a contribution does not satisfy the thermodynamic requirements necessary for representation as reversible heat exchange between neighboring equilibrium states. Therefore no corresponding reversible heat increment δQ_{rev} exists for this contribution.

If this non-Clausius contribution were nevertheless inserted into the entropy flux through the classical Clausius relation, the resulting entropy transfer would no longer possess a reversible thermodynamic interpretation. This contradicts the defining assumption underlying the Clausius entropy transfer.

Hence, only Clausius-effective heat may appear explicitly in the entropy flux and entropy supply terms of the local entropy inequality, whereas non-Clausius energy contributions must remain exclusively within the internal-energy balance.

Theorem 1. demonstrates that the proposed decomposition is not introduced as an additional constitutive assumption but follows directly from the thermodynamic meaning of Clausius entropy transfer. The exclusion of non-Clausius energy from the entropy inequality is therefore a structural consequence of the classical definition of entropy rather than a modification of the second law itself. The proposed formulation preserves the classical interpretation of entropy while extending its applicability to systems involving energy-transfer mechanisms that cannot be represented as reversible heat exchange.

3.3. Relation to the energy balance

The local energy balance retains its standard form:

$$\rho\dot{e} = \boldsymbol{\sigma} : \nabla \boldsymbol{V} - \nabla.\boldsymbol{q}^{C} - \nabla.\boldsymbol{q}^{NC} + r^{C} + r^{NC} \tag{7}$$

Thus, all forms of energy transfer, Clausius and non-Clausius alike, are fully accounted for in the first law. The distinction introduced here affects only the application of the second law. This separation resolves the conceptual tension identified in Section 3 that states that the second law is no longer applied to quantities whose thermodynamic nature has not been established.

3.4. Basic example: viscous fluid with fourier heat conduction

In order to demonstrate the practical meaning of the Clausius-effective heat, consider an incompressible Newtonian fluid with Fourier heat conduction. In an adiabatic isolated viscous system, entropy increases because the dissipated mechanical energy is converted into internal energy, not because viscous dissipation is treated as entropy-producing heat in the Clausius sense. The total heat flux and internal dissipation are given by:

$$\boldsymbol{q}_{total} = -k\nabla T \tag{8-a}$$

$$\Phi = 2\mu \boldsymbol{D} : \boldsymbol{D} \tag{8-b}$$

Where k is the thermal conductivity, μ the viscosity and $\boldsymbol{D}$ the rate-of-deformation tensor. In the conventional thermodynamic treatment, the viscous dissipation rate Φ is associated with irreversible mechanical conversion and contributes to the entropy production of the system through the complete thermodynamic balance. It should therefore not be interpreted as a heat flux merely because the mechanical energy is ultimately converted into internal energy. Within the present structural classification, Φ is regarded as a non-Clausius volumetric energy contribution: it remains fully present in the internal-energy balance, but it is not identified with a reversible thermal exchange across the system boundary.

We thus decompose the volumetric heating term as:

$$r^{C} = 0, \quad r^{NC} = \Phi$$

The Fourier heat flux is classified here as Clausius-effective because it represents a thermal energy-transfer pathway that admits the required reversible reference process for the entropy-transfer term. This classification does not imply that finite-gradient Fourier conduction is itself a reversible process; rather, the Clausius-effective designation refers to the thermodynamic character of the energy-transfer channel and its admissibility in the entropy-transfer representation:

$$\boldsymbol{q}^{C} = -k\nabla T, \quad \boldsymbol{q}^{NC} = 0$$

Substituting these expressions into Eq. (6) gives the revised entropy inequality

$$\rho\dot{S} \geq -\nabla.\left(\frac{\boldsymbol{q}^{C}}{T}\right) = -\nabla.\left(\frac{-k\nabla T}{T}\right)$$

Notably, the viscous dissipation term Φ does not appear as a Clausius entropy-transfer contribution in the revised inequality. Its energetic effect remains fully present in the internal-energy balance, while its irreversible thermodynamic consequences are represented through the appropriate entropy-production structure of the complete system. Thus, the present formulation does not eliminate viscous entropy generation; it distinguishes irreversible mechanical energy conversion from boundary heat transfer that carries entropy in the Clausius sense.

The classical entropy inequality is recovered only if one makes the additional assumption $r^{C} = \Phi$, which the present work shows to be thermodynamically unjustified.

It is important to emphasize that viscous dissipation is not eliminated from the thermodynamic description. Rather, it is excluded only from the Clausius-effective heat contribution because it does not correspond to a reversible thermal transfer. Its effect remains fully present in the

internal-energy balance and is manifested through temperature rise and irreversible mechanical degradation. Consequently, the proposed entropy inequality does not deny entropy generation in viscous systems; it only prevents mechanically mediated dissipation from being misidentified as Clausius heat.

This example demonstrates that the proposed decomposition naturally separates reversible thermodynamic heat from mechanically mediated, non-Clausius energy transfers, without altering the form of the first law.

3.5. Compatibility with the Coleman-Noll procedure

The classical Coleman-Noll argument determines thermodynamic restrictions by inserting the entropy inequality into the energy balance and requiring the resulting quadratic form in the constitutive variables to be non-negative for all admissible processes.

Using the decomposed heat flux $\boldsymbol{q} = \boldsymbol{q}^{C} + \boldsymbol{q}^{NC}$ and inserting the revised inequality (6) into the internal energy balance (7), one obtains:

$$\rho\dot{e} - \rho T\dot{S} \leq \boldsymbol{\sigma} : \nabla \boldsymbol{V} - \nabla . \boldsymbol{q}^{NC} + r^{NC} \tag{9}$$

The non-Clausius terms identified in this formulation are not interpreted as Clausius entropy-transfer contributions. Their presence in the energy balance nevertheless affects the thermodynamic evolution of the system and may be associated with irreversible entropy production through the complete dissipation structure. Accordingly, the revised classification should not be interpreted as removing these mechanisms from the second law; it separates their energetic role from their representation as boundary entropy transfer.

As a result, the reduced dissipation inequality to be enforced on constitutive equations becomes:

$$\boldsymbol{\sigma} : \nabla \boldsymbol{V} - \nabla . \boldsymbol{q}^{NC} + r^{NC} \geq 0 \tag{10}$$

Eq. (10) represents a reduced form of the local dissipation inequality that results from separating Clausius-effective and non-Clausius contributions to energy transfer. In this representation, only the Clausius-effective heat flux and heat supply participate in the entropy balance, while the non-Clausius components are represented outside the Clausius entropy-transfer term and enter the corresponding dissipation structure of the constitutive formulation. Consequently, the terms $\boldsymbol{q}^{NC}$ and r^{NC} are not associated with thermal conjugate variables in the entropy relation. Instead, they represent energy transfer mechanisms that contribute to dissipation without constituting thermodynamically admissible heat exchange in the Clausius sense. This separation clarifies the structural origin of dissipation and prevents the misinterpretation of all dissipative processes as heat in the classical formulation. This also shows that the Clausius-effective terms $\boldsymbol{q}^{C}$ and r^{C} carry the thermodynamic entropy structure, while the non-Clausius components are not assigned a direct Clausius entropy-transfer term in the present formulation. Their constitutive influence, including possible coupling to temperature, internal variables, or other thermodynamic fields, must nevertheless be determined from the complete constitutive structure of the particular material. Thus, the proposed classification does not by itself imply that non-Clausius mechanisms impose no thermodynamic constraints; it specifies only that they are not represented as reversible heat in the entropy-transfer term.

Thus the Coleman-Noll procedure remains mathematically intact, with the revised inequality providing a cleaner separation between thermal and non-thermal energy contributions.

3.6. Uniqueness of the Clausius-effective decomposition

The proposed formulation decomposes the total heat interaction into two components,

$$\delta Q = \delta Q^{C} + \delta Q^{NC}$$

Where δQ^{C} denotes the Clausius-effective heat responsible for entropy transfer across the system boundary, whereas δQ^{NC} represents energetic contributions that do not directly transport entropy across the boundary.

An important question is whether this decomposition is unique or merely one possible partition of the total heat interaction.

The uniqueness follows directly from the defining property of Clausius-effective heat.

Theorem 2.

Suppose two admissible decompositions exist,

$$\delta Q = \delta Q^{C} + \delta Q^{NC}$$

And:

$$\delta Q = \delta Q^{'C} + \delta Q^{'NC}$$

Both decompositions satisfy the defining requirement that the Clausius-effective component is the unique part of the energy exchange responsible for entropy transfer across the system boundary.

Proof.

Assume:

$$\delta Q^{C} \neq \delta Q^{'C}$$

Then one of the two decompositions necessarily attributes part of the entropy-transfer contribution to the non-Clausius component, or conversely assigns an entropy-neutral contribution to the Clausius-effective component.

Both possibilities contradict the defining criterion of the decomposition.

Therefore:

$$\delta Q^{C} = \delta Q^{'C}$$

and consequently:

$$\delta Q^{NC} = \delta Q^{'NC}$$

Hence, the proposed decomposition is unique.

4. Interpretation and characterization of non-Clausius heat

The reduced dissipation inequality (10) isolates the contribution of non-Clausius terms to mechanical dissipation, making it natural to clarify the physical nature of such terms. To provide a concrete interpretation of the quantities $\boldsymbol{q}^{NC}$ and r^{NC} appearing in Eq. (10), we now characterize the thermodynamic meaning and structural role of non-Clausius energy transfer. The notion of non-Clausius heat may initially appear abstract. In this section, we clarify its physical meaning and discuss possible ways to characterize it.

4.1. Physical interpretation

Non-Clausius heat refers to forms of energy transfer that cannot be identified with a reversible thermal interaction or mapped to an equivalent heat exchange at a well-defined temperature. Accordingly, such contributions are not represented as Clausius entropy-transfer terms merely on the basis of their energetic magnitude. This classification does not imply that the associated physical processes are entropy-neutral; irreversible processes may still contribute to internal entropy production through the complete thermodynamic evolution of the system.

Typical manifestations include energy dissipation arising from internal friction or inelastic deformation, energetically significant rearrangements of internal structure, and mechanically induced microscopic agitation that lacks a reversible thermal counterpart. In complex or active materials, externally powered or self-driven processes may also give rise to similar non-Clausius channels of energy redistribution. Although such energy transfers may ultimately lead to observable

temperature changes, their thermodynamic pathways are not equivalent to Clausius heat exchange in the sense of reversible thermal interaction.

It is important to emphasize that the present framework does not invalidate the conventional entropy production concept used in continuum thermodynamics. Rather, it clarifies that the classical formulation effectively assumes all dissipative energy transfers to be thermodynamically equivalent to Clausius heat. The revised approach makes this assumption explicit and relaxes it, thereby restricting the association of entropy production to those energy exchanges that admit a reversible thermodynamic reference. In this limiting sense, the classical entropy inequality is recovered as a special case of the present, more structurally refined formulation.

4.2. Experimental identification of Clausius-effective heat

The Clausius-effective heat introduced in the present formulation is not regarded as an independently measurable thermodynamic variable.

Instead, it is identified through the combined application of energy conservation, boundary energy fluxes, and entropy balance.

Experimentally, the total energy exchange is first determined using conventional calorimetric or energy-balance measurements.

The entropy balance then separates the portion of this energy that contributes to entropy transfer across the system boundary from the portion associated with irreversible internal processes.

Consequently, the Clausius-effective heat should be interpreted as an inferred thermodynamic quantity rather than a directly measured observable, in the same sense that entropy production itself is generally evaluated from thermodynamic balances rather than measured independently.

This interpretation provides an operational procedure for identifying the Clausius-effective contribution without introducing additional measurable state variables.

4.3. General procedure for identifying clausius-effective and non-Clausius energy

The structural decomposition introduced above becomes operational only when a specific material, process, and set of state variables are specified. Accordingly, the identification of Clausius-effective and non-Clausius contributions should not be regarded as a universal constitutive formula. Rather, it can be implemented through a process-dependent thermodynamic procedure. The following steps provide a general framework.

First, all energetic contributions must be identified from the complete internal-energy balance. This includes boundary energy fluxes, volumetric energy supplies, mechanical power, chemical contributions, electromagnetic energy transfer, microstructural energy storage, and other relevant mechanisms. At this stage, no energetic contribution is excluded from the first law.

Second, each energy-transfer mechanism should be classified according to whether it represents a thermal interaction capable of being associated with entropy transfer across the system boundary. A contribution is a candidate for the Clausius-effective part if a reversible reference process can be defined in which the corresponding energy exchange can be represented as reversible heat at the relevant thermodynamic temperature. Contributions that represent mechanical work, irreversible deformation, internal structural rearrangement, or other non-thermal energy pathways are not automatically assigned to the Clausius-effective part.

Third, the reversible and irreversible portions of mechanically mediated energy transfer should be distinguished where appropriate. For example, the stress power may be decomposed schematically into a reversible part associated with recoverable thermoelastic or elastic energy storage and an irreversible part associated with viscous, plastic, or other dissipative mechanisms. The latter remains present in the energy balance but is not thereby converted into a Clausius heat flux. This step provides a practical connection between the proposed structural criterion and established thermodynamic decompositions of reversible and dissipative processes.

For a thermomechanically coupled continuum, the mechanical power may be represented schematically as:

$$\boldsymbol{\sigma} : \nabla \boldsymbol{V} = P^{rev} + P^{irr}$$

Where P^{rev} denotes the recoverable part associated with elastic or thermoelastic energy storage, while P^{irr} represents irreversible mechanical dissipation, such as viscous or plastic dissipation. This decomposition is schematic and process-dependent rather than a universal constitutive identity. Its role in the present framework is diagnostic: P^{irr} is not converted into a Clausius heat flux merely because it ultimately contributes to internal energy, whereas a thermal exchange associated with a reversible reference process may be classified as Clausius-effective.

Fourth, the energetic variables should be examined with respect to their thermodynamic independence. If the internal energy is represented in terms of independent and dependent components:

$$e = e^{ind} + e^{dep}$$

The rates of the independent components can be used to identify the energetic pathways that are candidates for an independent reversible thermodynamic exchange. Dependent components are not automatically assigned a separate entropy-transfer contribution merely because their energy changes during the process. Their role must instead be assessed through the constitutive relations and constraints of the particular system.

Fifth, after the candidate Clausius-effective contributions have been identified, the total energy transfer can be partitioned according to:

$$\boldsymbol{q}^C = \boldsymbol{q}_{total} - \boldsymbol{q}^{NC}, \quad r^C = r_{total} - r^{NC}$$

With the important requirement that the two sides of each relation have the same physical dimensions and represent the same type of energetic quantity. In particular, a volumetric dissipative power cannot be directly subtracted from a heat-flux vector; fluxes and volumetric sources must therefore be classified separately.

Finally, the proposed classification should be checked against the entropy balance. The resulting Clausius-effective contribution must provide a thermodynamically consistent entropy-transfer term, while the remaining non-Clausius mechanisms must be accounted for through the appropriate irreversible processes, internal entropy production, or other terms required by the complete thermodynamic formulation. This final consistency check prevents the structural decomposition from becoming an arbitrary mathematical partition.

The procedure may be summarized as

$$\begin{aligned} &\textit{Energy balance} \rightarrow \textit{Identify mechanisms} \rightarrow \textit{Separate thermal and non} \\ &\quad - \textit{thermal pathways} \rightarrow \textit{Identify reversible reference process} \\ &\quad \rightarrow \textit{Classify } \boldsymbol{q}^C, r^C \rightarrow \textit{Check entropy balance} \end{aligned}$$

The same procedure can, in principle, be applied to different classes of continua. In plasticity, for example, recoverable elastic energy and plastic dissipation should be distinguished before assigning an energy contribution to the Clausius-effective sector. In phase-field or microstructural models, gradient and interfacial energies may evolve during the process, but their energetic evolution does not by itself establish a Clausius heat contribution; the associated reversible and dissipative mechanisms must be identified separately. In electromagnetically coupled systems, electromagnetic work and energy transport likewise require distinction from genuine thermal exchange before an entropy-transfer contribution is assigned. Thus, the proposed framework is intended as a thermodynamic classification procedure rather than as a material-specific constitutive equation.

4.4. Experimental strategies for distinguishing energy-transfer pathways

Although the present framework is formulated theoretically, its central distinction can in principle be examined experimentally by combining independent measurements of energy transfer, mechanical work, temperature evolution, and entropy-related quantities. The key requirement is to avoid inferring entropy transfer solely from the total amount of energy entering or leaving a system.

A useful experimental strategy is to perform a controlled reference experiment in which the same initial and final thermodynamic states are connected through a process designed to minimize irreversible mechanisms. The energy exchange associated with this reference process provides a basis for identifying the portion that can admit a reversible Clausius representation. A second experiment can then be performed under otherwise comparable conditions in which a selected irreversible mechanism, such as viscous dissipation, plastic deformation, friction, or internal structural relaxation, is activated. Calorimetric measurements, mechanical work measurements, heat-flux measurements, and temperature-field measurements can then be combined with the complete energy balance to determine how the total exchanged energy is partitioned among thermal and non-thermal pathways.

For a viscous system, for example, the mechanical power supplied to the system and the resulting viscous dissipation can be measured independently from the boundary heat flux. If the system is thermally isolated, a temperature increase may occur despite the absence of an imposed external heat flux. This observation demonstrates that an increase in internal energy does not by itself establish that the corresponding energy transfer should be represented as a Clausius heat flux. The proposed framework would classify the externally imposed thermal exchange and the mechanically generated dissipative energy according to their distinct thermodynamic pathways rather than treating both as the same type of entropy-carrying energy transfer.

A similar strategy can be applied to plastic or microstructural processes. Mechanical loading and unloading can be measured together with temperature and heat-flow responses. The recoverable energetic contribution can be identified from the reversible loading–unloading response, whereas hysteresis and irreversible mechanical work provide information about dissipative pathways. The comparison between the measured energy balance and the entropy balance can then be used to assess which energy exchanges admit a reversible thermal representation.

The experimentally relevant distinction is therefore not necessarily a difference in the total entropy production predicted by two formulations. Rather, the proposed framework provides a criterion for distinguishing entropy transfer across the boundary from irreversible entropy production associated with internal processes. In systems where these mechanisms are strongly coupled, measurements that independently resolve heat transfer, mechanical work, and internal energy storage can therefore test whether an energy contribution has been incorrectly assigned a Clausius interpretation.

More generally, a potential experimental test of the framework consists of the following sequence:

$$\begin{aligned}&\textit{Measure energy exchange} \rightarrow \textit{Resolve thermal and non}-\textit{thermal pathways}\\ &\qquad\qquad\qquad\qquad \rightarrow \textit{Identify a reversible reference process}\\ &\qquad\qquad\qquad\qquad \rightarrow \textit{Evaluate entropy transfer}\\ &\qquad\qquad\qquad\qquad \rightarrow \textit{Compare with internal entropy production}.\end{aligned}$$

Such experiments would not test the validity of the second law itself. Rather, they would test the proposed structural classification of exchanged energy and determine whether a particular energy-transfer pathway can consistently be associated with a Clausius entropy-transfer contribution.

4.5. Relation to system-surroundings entropy balances

The proposed distinction between Clausius-effective and non-Clausius energy transfers should be interpreted within, rather than separately from, the standard system–surroundings formulation of the second law. For a thermodynamic system, the entropy balance may be written schematically as:

$$\frac{dS_{sys}}{dt} = \dot{S}_{in} - \dot{S}_{out} + \dot{S}_{gen}, \dot{S}_{gen} \geq 0$$

where the first two terms describe entropy transfer across the system boundary and $\dot{S}_{gen}$ denotes internal entropy production. In the classical Clausius representation, heat transfer across a boundary contributes to entropy transfer through a term of the form $\frac{\delta Q}{T}$, provided that the corresponding thermal exchange admits the appropriate thermodynamic interpretation.

Within this framework, the present decomposition does not introduce a second entropy balance. Rather, it provides a structural criterion for identifying which portion of an energy exchange is to be associated with the entropy-transfer term. The Clausius-effective contribution δQ^{C} corresponds to the portion of boundary energy exchange that admits a reversible thermal representation and can therefore enter the entropy-transfer term. The non-Clausius contribution δQ^{NC}, in contrast, is not assigned an entropy-transfer term merely because it appears in the internal-energy balance.

This distinction is important because energy transfer and entropy transfer are not generally interchangeable concepts. Mechanical work, for example, can cross the system boundary without being represented as a heat contribution $\frac{\delta Q}{T}$. Similarly, internally generated dissipative energy may alter the internal-energy state of the system without constituting an entropy flux across its boundary. Such processes can nevertheless be associated with irreversible entropy production through the complete thermodynamic evolution of the system. Thus, exclusion of δQ^{NC} from the Clausius entropy-transfer term should not be interpreted as exclusion of the associated process from the second law or as a claim that entropy cannot be produced by irreversible processes.

The same distinction applies to the surroundings. A non-Clausius energy transfer is not, by definition, an entropy-neutral quantity in every possible interaction. Its effect on the entropy of the surroundings depends on the physical mechanism by which the energy is exchanged. If the exchange occurs as work or another non-thermal energy pathway, it does not acquire an entropy-transfer contribution simply from the amount of energy transferred. If the same overall process subsequently involves irreversible thermalization, entropy changes may occur in the system, the surroundings, or both. Consequently, the present criterion classifies the thermodynamic character of the energy-transfer pathway rather than asserting that non-Clausius energy has no possible influence on entropy elsewhere.

This interpretation is consistent with the broader distinction in non-equilibrium thermodynamics between energy transfer, entropy transfer, and internal entropy production. Related discussions of energy-stable thermodynamic formulations and dissipative contributions provide a useful conceptual connection between the present structural distinction and established system-level entropy balances [44].

4.6. Predictive consequences of the revised entropy inequality

The proposed entropy inequality does not merely introduce a different mathematical notation.

Instead, it provides a different physical attribution of entropy transfer and entropy generation.

Under the classical formulation,

$$dS \geq \frac{\delta Q}{T}$$

the entire energetic interaction is conventionally represented by a single heat term.

In contrast, the present formulation distinguishes between the entropy-carrying component, δQ^C, and energetic contributions that remain entirely within the irreversible entropy-production mechanism.

Consequently, the revised inequality predicts the same global compliance with the second law while providing a different physical interpretation of irreversible processes.

For systems involving viscous dissipation, frictional heating, or other internally generated energy conversion mechanisms, the proposed formulation predicts that only the Clausius-effective part contributes to entropy transfer across the boundary, whereas internally generated irreversible energy contributes exclusively to entropy production.

Although both formulations satisfy the second law, they differ in the physical allocation of entropy transfer and entropy generation.

This distinction becomes particularly relevant in systems where substantial irreversible internal energy conversion occurs.

4.7. Quantifying Clausius-effective heat

The present framework does not prescribe a unique method for determining $\boldsymbol{q}^C$ and r^C. Instead, it establishes a conceptual constraint. Based on that, only energy transfer that admits a reversible reference process may enter the entropy inequality. To make this conceptual criterion more operational, a general identification procedure was introduced in Section 4.3.

In practical modeling, this may be achieved by:

1. Identifying reversible thermal interaction mechanisms explicitly
2. Subtracting known dissipative or structurally mediated energy transfers from the total heat
3. Or introducing constitutive relations that directly specify the Clausius-effective heat flux

The precise form of this decomposition is expected to be material and process-dependent. Importantly, the proposed framework allows such distinctions to be introduced systematically, rather than implicitly and inconsistently.

For accurate modeling, it is crucial to identify which energy components are dynamically independent, as only these contribute to the Clausius-effective heat. Non-independent or purely dependent components, while present in the energy balance, should be excluded from the entropy calculation.

Definition 1. (Clausius-effective energy).

An exchanged energy contribution is called Clausius-effective if there exists a reversible reference process connecting the same equilibrium states for which the exchanged energy is representable as the reversible heat appearing in the Clausius relation.

Theorem 3. (Necessary and Sufficient Condition for Entropy Admissibility).

Let E denote an arbitrary exchanged energy contribution appearing in the local internal-energy balance. Then EEE contributes explicitly to the local entropy inequality if and only if it admits a reversible thermodynamic representation in the Clausius sense; namely, there exists a reversible reference process for which

$$dS = \frac{\delta Q_{rev}}{T}$$

with E identified as the corresponding reversible heat transfer.

Consequently,

$$E \in C \quad \Leftrightarrow \quad E \text{ is Clausius} - \text{effective}$$

Proof.

(Necessity) Assume that an exchanged energy contribution enters explicitly into the entropy flux or entropy supply terms of the local entropy inequality.

Since these terms originate from the Clausius definition of entropy transfer, every contributing energy component must admit representation as reversible heat exchange divided by the absolute temperature.

Therefore, a reversible reference process necessarily exists for which

$$dS = \frac{\delta Q_{rev}}{T}$$

establishing that the exchanged energy is Clausius-effective.

(Sufficiency) Conversely, assume that an exchanged energy contribution possesses a reversible Clausius representation.

Then its associated entropy transfer is uniquely defined by

$$dS = \frac{\delta Q_{rev}}{T}$$

and the corresponding energy contribution may consistently appear within the entropy flux and entropy supply terms of the local entropy inequality.

Therefore, every Clausius-effective energy contribution is thermodynamically admissible in the entropy balance.

Combining the necessity and sufficiency parts establishes the equivalence.

Theorem 3. establishes a precise thermodynamic criterion governing the admissibility of exchanged energy in the local entropy inequality. Rather than relying on constitutive assumptions or phenomenological classifications, the proposed criterion follows directly from the Clausius definition of entropy transfer. The distinction between Clausius-effective and non-Clausius energy therefore becomes a mathematically well-defined property instead of a heuristic interpretation. The revised entropy inequality consequently acquires a rigorous selection principle identifying which exchanged energy contributions may consistently participate in entropy transport.

5. Consistency with advanced thermodynamic frameworks

The revised entropy inequality proposed in this work is fully consistent with the structure and objectives of modern non-equilibrium thermodynamics. Contemporary frameworks frequently distinguish between multiple internal reservoirs, microstructural storage modes, and energetically significant but non-thermal pathways, emphasizing that entropy production arises from irreversible internal processes rather than from arbitrary forms of energy transfer. Within these approaches, heat is treated not as a universal label for every energy flux appearing in the balance equations, but as a specific mode of reversible thermal exchange that can be meaningfully compared to a reference process at a well-defined temperature. The present formulation aligns naturally with this viewpoint by enforcing that only Clausius-effective energy transfers enter the entropy inequality. This clarification resolves conceptual ambiguities that typically arise in strongly coupled continua, where mechanical, chemical, or structural energy flows may be present but lack a reversible thermal interpretation.

Viewed from a broader perspective, the proposed selective-entropy formulation acts as a refinement of the second law that preserves its classical mathematical structure while extending its conceptual validity. By structurally separating Clausius-effective and non-Clausius heat, the theory prevents the over-interpretation of entropy production in systems where mechanically mediated or non-thermal energy exchanges dominate, and it provides a clearer thermodynamic foundation for constitutive modeling in materials exhibiting internal dissipation, microstructural evolution, or active behavior. This distinction is particularly important for systems operating far from equilibrium, for which reversible thermal pathways coexist with irreversible structural mechanisms.

Beyond its immediate theoretical implications, the framework suggests promising avenues for future research. Constitutive models may be developed in which Clausius-effective heat is treated as an explicit state-dependent field. Multi-physics systems, such as those involving electromagnetism, chemical reactions, or phase transformations, may benefit from a decomposition that identifies their reversible thermal contributions, and also experimental strategies can be designed to differentiate between thermally reversible and thermally irreversible channels of energy exchange. Collectively, these perspectives show that the selective-entropy formulation is not an isolated modification of the entropy inequality, but rather part of a broader synthesis that brings continuum thermodynamics into closer alignment with contemporary views of irreversibility and internal energy structure.

6. Structural quantification of clausius-effective and non-Clausius energy transfers

6.1. Structural decomposition and thermodynamic interpretation

The distinction between Clausius-effective and non-Clausius energy transfers is introduced here not as an operational measurement prescription but as a structural consequence of the entropy definition itself. Even in situations where an explicit experimental separation is nontrivial, the decomposition acts as a conceptual constraint that preserves thermodynamic consistency by preventing the indiscriminate attribution of entropy production to all channels of energy redistribution. When the total heat flux and volumetric energy sources are expressed in decomposed form, the first law remains strictly intact - energy is fully conserved - while the second law selectively constrains only those components of energy transfer that admit a reversible thermal interpretation. This separation does not rely on constitutive assumptions but instead reflects the fundamental requirement that entropy production be associated exclusively with thermodynamically meaningful heat exchange.

Within this structural framework, the Clausius-effective contributions are identified through the relations:

$$\boldsymbol{q}^{C} = \boldsymbol{q}_{total} - \boldsymbol{q}^{NC}, \quad r^{C} = r_{total} - r^{NC} \tag{11}$$

Which clarify the portion of energy transfer relevant to the entropy inequality without modifying the underlying energy balance. The decomposition is therefore structural rather than material-specific. In fact, it does not prescribe how energy is transported, but rather how different transfer mechanisms must be classified with respect to the second law. In particular, only those contributions that can, in principle, be traced back to a reversible thermal pathway are permitted to enter the entropy balance, whereas intrinsically irreversible mechanisms - such as viscous dissipation, internal structural rearrangements, or chemically driven energy release - remain confined to the internal energy equation.

The practical implementation of this separation depends on the dominant mechanisms of energy transfer in the process under consideration, yet the underlying criterion remains universal. In solid materials governed by conduction-dominated transport, the presence of a well-defined temperature field allows most of the energy flux to be naturally interpreted as Clausius-effective, with non-Clausius contributions arising only in non-ideal situations involving internal dissipation or microstructural activity. In contrast, in viscous fluids, mechanically mediated energy transfer plays a central role. In fact, the stress power associated with viscous deformation generates internal energy through pathways that lack any reversible thermal counterpart. As a result, viscous dissipation is classified within the non-Clausius sector in the present framework, while only the remaining thermally mediated part of the energy transfer qualifies as Clausius-effective. Also, note that although viscous dissipation increases the internal energy and may raise the temperature, this temperature rise is not a result of Clausius-effective heat transfer but is generated internally through a mechanically mediated mechanism.

For the viscous-fluid example considered in Section 3, the thermal heat flux and viscous dissipation must be treated as distinct energetic quantities with different physical dimensions. The appropriate structural classification is therefore:

$$\boldsymbol{q}^{C} = \boldsymbol{q}_{total}, \quad \boldsymbol{q}^{NC} = 0 \tag{12}$$

For the heat-flux sector, while the volumetric energy supply is decomposed as:

$$r^{C} = 0, \quad r^{NC} = \Phi$$

Thus, the viscous dissipation rate is retained as a volumetric non-Clausius energy contribution and is not subtracted from the heat-flux vector. More generally, flux contributions and volumetric source contributions must be decomposed separately according to their physical nature and dimensions. The relations above are specific to the idealized viscous-fluid example and should not be interpreted as universal constitutive identities.

This expression is not a universal identity but exemplifies the core message of the present framework. In fact, not every contribution appearing in the energy balance constitutes thermodynamic heat in the Clausius sense. It is important to note that the presence of mechanical dissipation in Eq. (12) does not imply that the associated energy loss must be interpreted as Clausius heat. Classical formulations often conflate irreversible dissipation with a corresponding heat input in the entropy balance, but this identification is not structurally justified. Within the present framework, the dissipative power generated by viscous stresses contributes to the non-Clausius sector of the energy transfer. Such contributions increase entropy only indirectly through their effect on the mechanical structure of the continuum, not through a thermodynamically admissible heat flux. This distinction clarifies that mechanical dissipation does not automatically induce a Clausius-effective heat flux and prevents the systematic over-identification of dissipative mechanisms as heat in the traditional continuum thermodynamics formulation.

A similar structural distinction arises in chemically active systems, where reactions introduce additional channels of energy exchange that may or may not be recoverable through a reversible thermal process. Depending on the reaction mechanism, a portion of the released or absorbed energy may be transferable to a thermal reservoir in a reversible manner, while the remainder is inherently irreversible and therefore non-Clausius. The Clausius-effective heat is thus determined by the fraction of reaction energy that admits a reversible thermal interpretation, with all other reaction-driven contributions assigned to the non-Clausius sector. In this sense, the total heat and volumetric energy sources can always be reorganized according to:

$$\boldsymbol{q}^{C} = \boldsymbol{q}_{total} - \boldsymbol{q}^{NC}, \quad r^{C} = r_{total} - r^{NC} \tag{13}$$

Ensuring that only Clausius-effective terms appear in the entropy inequality, while non-Clausius transfers remain confined to the internal energy balance.

Applying this structural decomposition to the local entropy inequality yields:

$$\rho\dot{S} \geq -\nabla\cdot\left(\frac{\boldsymbol{q}_{total} - \boldsymbol{q}^{NC}}{T}\right) + \frac{r_{total} - r^{NC}}{T} \tag{14}$$

Which provides a process-aware yet conceptually unified method for enforcing the second law across a wide range of continuum systems. The resulting formulation bridges the conceptual separation introduced in the preceding sections with concrete computation, while maintaining a clear distinction between thermodynamically admissible heat transfer and structurally non-Clausius energy exchange.

6.2. Role of independent energy components in entropy

Considering the energy structure evolution equations [9–11], the internal energy of a continuum can be formally decomposed into a set of independent energy components, e_i^{ind}, and dependent components arising from internal couplings or constraints. Independent energetic components provide natural candidates for identifying distinct reversible thermodynamic exchange pathways, whereas dependent components must be assessed through the constitutive constraints and coupling relations of the specific system. Formally, we can write:

$$\sum_i \dot{e}_i^{ind} + \sum_j \dot{e}_j^{dep} = \dot{e} \tag{15}$$

Where $\dot{e}_i^{ind}$ are dynamically active and $\dot{e}_j^{dep}$ do not independently affect the thermodynamic state. Consequently, the Clausius-effective heat may be represented primarily in terms of the independent components, subject to the constitutive constraints and coupling relations of the system:

$$\boldsymbol{q}^C = g\left(\dot{e}_i^{ind}\right), \quad r^C = f\left(\dot{e}_i^{ind}\right) \tag{16}$$

A structurally consistent form of the Clausius-effective flux and volumetric source may then be written as:

$$g\left(\dot{e}_i^{ind}\right) = \sum_i A_i \dot{e}_i^{ind}, f\left(\dot{e}_i^{ind}\right) = \sum_i B_i \dot{e}_i^{ind} \tag{17}$$

Where A_i and B_i are material-dependent coefficients whose values are determined by the reversible portion of the underlying energy structure. It is important to emphasize that the coefficients A_i and B_i introduced in Eq. (17) do not represent thermodynamic forces in the classical sense. Rather, they should be interpreted as structural weighting factors that project the contribution of dynamically independent energy components onto the Clausius-effective heat flux and volumetric heat supply. In this sense, the coefficients characterize how reversible portions of the underlying energy structure participate in thermodynamically admissible heat exchange. Their values are therefore determined by the reversible sector of the constitutive response and may depend on the material class and the dominant transport mechanisms. Importantly, this representation does not impose a specific constitutive model but provides a general structural form that ensures that dynamically independent energy components provide a natural basis for identifying contributions to Clausius-effective heat, while the role of dependent components must be determined from the constitutive and coupling structure of the system. This establishes that the Clausius-effective heat is not a phenomenological separation but a structural consequence of the underlying dynamical decomposition of internal energy. So, Eq. (14) can be rewritten as follows:

$$\rho \dot{S} \geq -\nabla \cdot \left(\frac{\sum_i A_i \dot{e}_i^{ind}}{T} \right) + \frac{\sum_i B_i \dot{e}_i^{ind}}{T} \tag{18}$$

Which represents a fully general but explicit realization of the functions f and g. This construction demonstrates that the selective contribution of independent components to entropy production is not merely conceptual, but admits a concrete mathematical representation without recourse to a specific constitutive model. This perspective provides a fundamental justification for the selective nature of Clausius-effective heat that states that energy transfers associated with dependent or constrained components are not assigned a separate Clausius entropy-transfer contribution solely on the basis of their energetic evolution; their thermodynamic role must be assessed from the complete constitutive structure. In practical terms, this principle guides the identification of reversible mechanisms and non-Clausius contributions in complex continua, especially those with strong internal coupling, dissipation, or active microstructure dynamics. Integrating this perspective into the revised entropy inequality ensures that the second law remains consistent with the underlying dynamical structure of the system.

7. Discussion

The principal outcome of the present formulation is the establishment of a structural selection principle governing the admissibility of exchanged energy in the local entropy inequality. Rather than assuming that every energy contribution appearing in the internal-energy balance necessarily participates in entropy transfer, the proposed framework identifies a precise thermodynamic criterion based on the existence of a reversible Clausius representation. Consequently, the distinction between Clausius-effective and non-Clausius energy is no longer phenomenological but follows directly from the thermodynamic structure of entropy itself.

The introduction of this admissibility criterion also clarifies the relationship between the local energy balance and the entropy inequality. The two balance laws remain fully consistent while serving fundamentally different thermodynamic roles: the internal-energy balance accounts for all exchanged energy irrespective of its physical origin, whereas the entropy inequality selectively incorporates only those contributions satisfying the Clausius definition of reversible heat transfer. This distinction provides a coherent interpretation of entropy transport in generalized thermodynamic systems involving multiple irreversible energy-transfer mechanisms.

The analysis developed here is not intended as a purely formal modification of the classical entropy inequality. Rather, the governing equations and representative examples, when considered in light of established principles from non-equilibrium thermodynamics and statistical physics, point toward a more comprehensive interpretation of energy exchange in continua. In particular, they suggest that a complete thermodynamic description must distinguish between the mere energetic presence of a term in the balance equations and its actual role in entropy-producing thermal transfer. In this sense, the present framework may be regarded as an analysis based on the evolutionary equations of energy structure: it seeks to identify which components of energy exchange are dynamically independent and thermodynamically effective in the Clausius sense, and which remain structurally real but non-entropic. The proposed decomposition therefore emerges not from a purely theory-dependent assumption, but from an attempt to place continuum balance laws in closer correspondence with the broader physical understanding of irreversible processes.

The results developed in this work reveal a structural limitation in the traditional formulation of the local entropy inequality. In the classical approach, all energy fluxes and volumetric sources appearing in the energy balance are automatically interpreted as thermodynamic heat, and therefore assumed to contribute to entropy production. This assumption is usually left implicit because, for a restricted class of materials, mainly simple thermoelastic media with Fourier conduction, it happens to yield correct predictions. However, once internal dissipation, structural rearrangements, or chemically driven energy exchanges are present, this identification becomes conceptually inconsistent. The present framework makes this distinction explicit by recognizing that only a subset of the exchanged energy admits a thermodynamic interpretation in the Clausius sense. This observation aligns with recent discussions on thermodynamic consistency in generalized continuum formulations, where improper identification of energy transfer mechanisms leads to structural inconsistencies in entropy production, as noted in [37].

It should be emphasized that the present analysis does not imply that the classical local entropy inequality is universally invalid. Rather, the point is that its standard form is strictly justified only when the energy transfers identified as heat in the balance law also admit a reversible Clausius interpretation. The proposed revision therefore acts as a structural qualification of the classical statement, intended for

generalized continua in which mechanically, structurally, or chemically mediated energy exchanges cannot be unambiguously interpreted as thermodynamic heat.

A key insight emerging from this study is that continuum systems possess multiple channels of energy redistribution, not all of which correspond to reversible thermal pathways. Mechanically mediated processes such as viscous dissipation, plastic flow, microstructural rearrangement, or reaction-driven energy release may alter the internal energy of a system without being interpretable as heat in a thermodynamic sense [29–34]. Classical continuum thermodynamics implicitly forces all such processes into the thermal category by inserting them into the entropy inequality. In contrast, the revised formulation introduced here treats these contributions as fundamentally non-Clausius. They remain fully represented within the first law but no longer contribute artificially to entropy production. This does not mean that viscous or other mechanically mediated dissipation is thermodynamically irrelevant. On the contrary, such processes remain fully active in the internal-energy balance and may produce temperature rise and irreversible material evolution. The present distinction only states that their contribution should not be automatically classified as Clausius-effective heat unless a reversible thermal pathway can be identified.

This distinction has significant implications for the modeling of complex continua. By separating Clausius-effective and non-Clausius pathways, the revised inequality prevents misclassification of mechanically generated energy as entropy-producing heat. This leads to a cleaner and more transparent interpretation of irreversibility, especially in systems where strong coupling between mechanical, thermal, and chemical processes obscures the actual thermodynamic structure. In simple systems in which the dominant energy-transfer pathway is conventional thermal exchange, the proposed formulation reduces to the classical description and may therefore be experimentally indistinguishable at the level of directly measured macroscopic observables. From a modeling standpoint, the distinction becomes operational whenever the energetic channels of a constitutive description can be separated into independent and dependent components. In that setting, the proposed framework provides a criterion for identifying which exchanges may consistently enter the entropy inequality and which must remain confined to the energy balance. The framework introduced here does not alter the mathematical machinery of continuum mechanics. Rather, it clarifies the physical meaning of the quantities entering the second law. In doing so, it aligns the continuum formulation more closely with modern thermodynamic theories that emphasize the role of reversible reference processes and the selective nature of entropy production.

Moreover, the distinction introduced in this work offers a unifying perspective across a wide range of materials and processes. It provides a natural interpretation for why some energy transfers generate entropy while others do not, and it clarifies longstanding ambiguities in the treatment of dissipative mechanisms. By grounding the Clausius-effective heat in the structure of independent energy components, the framework is inherently compatible with contemporary theories [7–11]. These theories suggest that entropy production arises only from dynamically relevant energy channels, a viewpoint that is fully consistent with the separation proposed in this study.

Overall, it must be noted that the contribution of this work is not merely a mathematical reformulation but a conceptual refinement. This interpretation echoes recent revisions of the phenomenological and statistical statements of the second law, which emphasize that entropy production must be associated only with thermodynamically admissible pathways [39–41]. It aims to improve the internal consistency between the first and second laws in continuum descriptions involving non-Clausius energy exchange. The implications extend beyond classical thermal problems and may influence the treatment of non-equilibrium systems, multi-physics couplings, and active materials.

7.1. Limitations of the defined problem

The problem considered in the present study is deliberately formulated at the level of local continuum thermodynamics. First, the analysis assumes that the continuum admits a meaningful local thermodynamic temperature and entropy description; microscopic nonequilibrium effects that cannot be represented through local continuum variables are outside the present formulation. Second, the proposed Clausius-effective/non-Clausius classification is a thermodynamic identification criterion rather than a material-independent constitutive law. Its practical implementation requires knowledge of the relevant energy-transfer mechanisms and constitutive constraints of the particular system. Finally, the viscous-fluid example is intended to demonstrate the structural distinction in a transparent setting and should not be interpreted as a universal constitutive proof for all dissipative materials. These limitations define the present scope while also identifying the principal directions in which the framework requires further theoretical, computational, and experimental development.

7.2. Limitations of the present formulation

The present formulation should be regarded as a structural reformulation of the entropy inequality rather than a replacement of classical thermodynamics.

The proposed decomposition relies on the identification of Clausius-effective heat as the unique entropy-carrying component of energy transfer.

Although this definition is mathematically well posed, further experimental investigations will be valuable for establishing practical identification procedures in complex thermodynamic systems.

Moreover, the present work has been developed within the framework of classical continuum thermodynamics.

Extensions toward non-equilibrium statistical mechanics, relativistic thermodynamics, quantum thermodynamics, and multi-scale transport processes remain open research directions.

Finally, the proposed formulation does not alter the validity of the second law itself.

Instead, it seeks to provide a more explicit structural separation between entropy transfer and irreversible entropy production.

8. Conclusions

The present study has examined the conventional local entropy inequality employed in continuum thermodynamics and has shown that its standard formulation implicitly assumes that every energy transfer appearing as heat in the internal-energy balance is thermodynamically admissible within the Clausius entropy relation. The analysis demonstrates that this assumption is not generally valid, particularly in generalized continua involving internal dissipation, structural rearrangement, multiphysics coupling, or other irreversible energy-transfer mechanisms for which no reversible Clausius interpretation exists.

A structural reformulation of the local entropy inequality has therefore been developed by distinguishing exchanged energy according to its thermodynamic admissibility rather than solely according to its energetic contribution. Central to this formulation is the establishment of a necessary and sufficient thermodynamic admissibility criterion demonstrating that an exchanged energy contribution participates explicitly in the entropy inequality if and only if it admits a reversible Clausius representation. This criterion provides a rigorous theoretical foundation for the proposed distinction between Clausius-effective and non-Clausius energy transfer, showing that the decomposition follows as a structural consequence of the thermodynamic definition of entropy transfer rather than as an additional constitutive assumption.

The proposed formulation preserves the first law of thermodynamics without modification and retains all exchanged energy within the local energy balance. Only the entropy-transfer term is restricted to Clausius-

effective contributions, while non-Clausius mechanisms may still contribute to internal entropy production through the complete thermodynamic evolution of the system. Consequently, mechanically or structurally mediated irreversible processes—including viscous dissipation, plastic deformation, microstructural evolution, chemical reaction pathways, or other strongly coupled energy-transfer mechanisms—remain fully represented within the energy balance while avoiding their automatic interpretation as Clausius heat solely because they contribute to internal energy.

The resulting framework therefore preserves the classical Clausius–Duhem inequality as a limiting case whenever all exchanged energy admits a reversible Clausius interpretation. For conventional thermoelastic continua governed by Fourier heat conduction, the classical and revised formulations become equivalent. The proposed formulation becomes relevant only for systems in which multiple energy-transfer mechanisms coexist and only a subset of those mechanisms satisfies the thermodynamic requirements associated with reversible heat exchange.

Beyond introducing a revised entropy inequality, the present formulation establishes a structural selection principle governing the admissibility of exchanged energy in entropy transport. This principle provides a consistent interpretation of the relationship between the local energy balance and the second law while strengthening the theoretical basis for constitutive modeling of generalized continua. The proposed framework is also consistent with the broader perspective of energy structure evolution equations, according to which entropy is associated only with dynamically independent energy components possessing thermodynamically meaningful reversible representations.

The principal contribution of the present study is therefore not merely a reformulation of the local entropy inequality but the establishment of a rigorous thermodynamic criterion identifying which exchanged energy contributions may consistently participate in entropy transport. This framework provides a clearer theoretical foundation for the analysis of dissipative, strongly coupled, and non-equilibrium continuum systems. Future investigations may focus on constitutive implementations, numerical simulations, and experimental studies capable of examining the consequences of the proposed admissibility criterion in complex thermodynamic processes.

Nomenclature

g_j	Value-dependent energy component with SI (metric) dimension (***Joule***).
h_n	Rate-dependent energy component with SI (metric) dimension (***Joule***).
δQ	Total heat exchange with SI (metric) dimension (***Joule***).
u_i	Energy component (Dependent and Independent) of the system with SI (metric) dimension (***Joule***).
T	Temperature with SI (metric) dimension (K°).
r	volumetric heat supply with SI (metric) dimension (W/m^3).
D	the rate-of-deformation tensor with SI (metric) dimension (**1/s**).
S	Specific entropy of the system with SI (metric) dimension ($\frac{Joule}{K\ kg}$).
V	Velocity field with SI (metric) dimension (**m/s**)
e	specific internal energy (**J/kg**)
q	heat flux (W/m^2)

Greek Letters

ε	Strain without SI (metric) dimension.
σ	Stress with SI (metric) dimension (pa).

CRediT authorship contribution statement

Saeed Shahsavari: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Data availability

No data associated in the manuscript.

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